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RESEARCH ARTICLE

# Ultra-Narrow Pure-Green MR-TADF Emitter with an FWHM of 12 nm Enables Superior-Performance Top-Emitting OLEDs with EQE Approaching 60%, Power Efficiency Over 300 lm W$^{-1}$, and CIE Coordinates of (0.14, 0.79)

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**Received:** 9 April 2026 | **Revised:** 5 June 2026 | **Accepted:** 30 June 2026

**Keywords:** BT.2020 | multi-resonance | organic light-emitting diodes | pure-green | thermally activated delayed fluorescence | ultra-narrow

## ABSTRACT
Achieving pure-green organic light-emitting diodes (OLEDs) with both precise spectral positioning and an ultra-narrow full-width at half-maximum (FWHM) remains highly challenging, as red-shifting emission into the pure-green regime is often accompanied by enhanced excited-state relaxation and spectral broadening. Herein, we report a molecular design strategy that reconciles green-region red-shift with intrinsic spectral narrowing through the synergistic integration of dibenzofuran fusion and cyano decoration within a *meta*-diboron framework. The proof-of-concept molecule DBFCN exhibits intrinsically ultra-pure green emission in dilute toluene, centered at 519 nm with a very small FWHM of 12 nm, placing it among the narrowest green multi-resonance thermally activated delayed fluorescence (MR-TADF) emitters reported so far. The bottom-emitting OLEDs deliver a maximum external quantum efficiency ($EQE_{max}$) of 38.5% and pure-green emission at 521 nm with an ultra-narrow FWHM of 13 nm. Owing to the intrinsically ultra-narrow emission and minimal spectral tail of DBFCN, photon loss in the top-emitting (TE) configuration is effectively mitigated, leading to markedly enhanced device performance. Consequently, the TE-device delivers a maximum power efficiency ($PE_{max}$) exceeding 300 lm W$^{-1}$ and a current efficiency ($CE_{max}$) of 232.5 cd A$^{-1}$, while meeting the green BT.2020 color gamut.

## 1 | Introduction

With ultra-high-definition (UHD) display technologies pushing relentlessly toward ultra-wide color gamut and improved energy utilization efficiency, luminescent materials that simultaneously deliver excellent luminous efficiency with high color purity are indispensable to next-generation organic light-emitting diode (OLED) displays [1–4]. Among the primary colors that define the gamut boundary, green emission critically determines the achievable coverage of the Broadcasting and Television 2020 (BT.2020) color gamut [5, 6]. An ideal pure-green emitter must simultaneously satisfy two stringent criteria: emission accurately located in the pure-green region and a highly compressed spectral width to maximize gamut utilization [7, 8]. Against this backdrop,

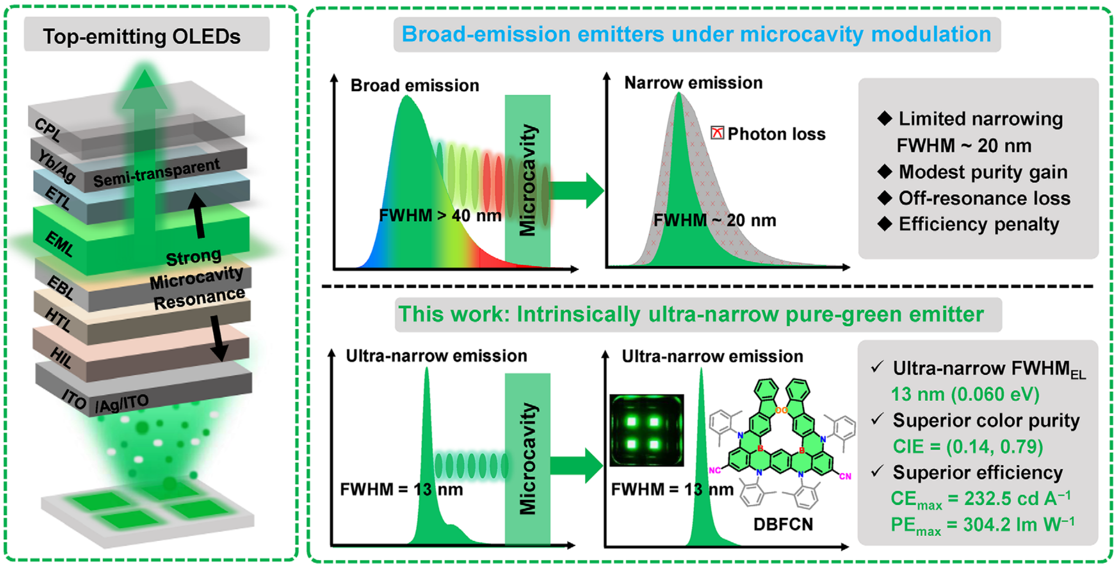


**FIGURE 1** | Light emission characteristics of top-emitting OLEDs based on green emitters with wide emission and intrinsic ultra-narrow emission.

multi-resonance thermally activated delayed fluorescence (MR-TADF) emitters, built on fused polycyclic skeletons that incorporate electron-donating and electron-withdrawing centers in an alternating arrangement, have emerged as compelling candidates for high-color-purity OLEDs by virtue of minimized excited-state structural relaxation and intrinsically narrowband emission [9–11]. The spectral sharpness is epitomized by *ν*-DABNA, established by Hatakeyama and co-workers, which remains the prototypical benchmark for ultra-narrow blue emission with a full-width at half-maximum (FWHM) of 14 nm [9]. Extending the ultra-narrow emission characteristic of the MR-TADF archetypes within the pure-green region is highly desirable for wide-color-gamut displays, yet remains exceptionally challenging [12–21]. In spite of the recent rapid development in green MR-TADF emitters, most reported systems still exhibit FWHM values exceeding 15 nm, underscoring spectral broadening during bathochromic shift as the central unresolved constraint on ultra-narrow pure-green emission [22–38]. In our previous work, cyano engineering of a *ν*-DABNA-derived framework preserved a remarkably small FWHM of 14 nm for green emission, underscoring the critical role of cyano decoration in restraining vibronic coupling and excited-state structural relaxation [13]. Nevertheless, the emission maximum was still offset from the ideal pure-green region, constraining further improvement in color purity. These findings indicate the central challenge is to drive MR-TADF emission to fall into the pure-green regime without compromising its spectral sharpness. This requirement becomes even more stringent in top-emitting (TE) OLEDs, which have been widely adopted in cutting-edge display panels owing to their high aperture ratio, directional light extraction, and microcavity-enabled spectral selectivity [39]. While the microcavity can improve color purity through selective enhancement around the cavity resonance, its effectiveness is critically determined by the intrinsic emission bandwidth of the emitter. For conventional emitters with broad emission, one of the primary functions of the microcavity is to serve as a passive spectral filter, selectively extracting resonant emission while suppressing off-resonance photons, thereby incurring an intrinsic penalty in photon utilization [40–42]. Consequently, the spectral purification achieved in TE-OLEDs often comes at the expense of limited net efficiency gain (Figure 1). This intrinsic mismatch underscores the challenge of developing pure-green emitters with genuinely narrowband emission and strong compatibility with microcavity modulation for next-generation TE-OLEDs.

Herein, we design a new molecule that deliberately couples $\pi$-extension with cyano decoration on a *meta*-diboron MR framework to reconcile red-shift toward the green region with intrinsic spectral narrowing. Specifically, incorporation of the dibenzofuran (DBF) unit into the framework narrows the energy gap ($E_g$) and shifts the emission toward the green region, whereas cyano moieties introduced para to the boron centers further refine the emission toward the pure-green target and simultaneously suppress excited-state structural relaxation. Such cooperative structural modulation enables the unusual yet highly desirable combination of red-shifted emission and further spectral narrowing. The resulting emitter, DBFCN, exhibits intrinsically ultra-narrow pure-green photoluminescence in dilute toluene, where the emission maximum is located at 519 nm, and the FWHM narrows to 12 nm (0.056 eV). This intrinsically ultra-narrow emission of DBFCN is well preserved in the corresponding electroluminescence, affording bottom-emitting OLEDs with sharply saturated pure-green emission at 521 nm, an ultra-narrow FWHM of 13 nm (0.060 eV), together with Commission Internationale de l'Eclairage (CIE) coordinates of (0.18, 0.75). Notably, its intrinsically narrowband emission is highly compatible with top-emitting microcavity architectures, allowing the spectral bandwidth to remain essentially unchanged while markedly alleviating off-resonance photon loss. Consequently, the TE-OLEDs deliver a high maximum power efficiency ($PE_{max}$) exceeding 300 lm W$^{-1}$ and a current efficiency ($CE_{max}$) of 232.5 cd A$^{-1}$, accompanied by excellent color purity with CIE coordinates of (0.14, 0.79), essentially meeting the BT.2020 green target. This work therefore establishes a viable molecular design strategy toward intrinsically ultra-narrow pure-green MR-TADF emitters and highlights the importance of compatibility with microcavity modulation in achieving high-performance TE-OLEDs for next-generation UHD displays.

## 2 | Results and Discussion

### 2.1 | Molecular Design and Theoretical Calculation

Figure S1 summarizes the molecular strategy and corresponding structures of DBN, DBFBN, and DBFCN. A central requirement for realizing highly pure-green MR-TADF emitters is to push the emission of blue *meta*-diboron MR skeleton to longer wavelengths without incurring the spectral broadening and tailing that typically accompany bathochromic tuning. With this objective in mind, we undertook a stepwise reconfiguration of the *meta*-diboron MR framework. Removal of the diphenylamine groups para to the boron atoms served as the initial electronic adjustment, enabling a moderate bathochromic shift while largely preserving the characteristic MR topology. The scaffold was then further fused with dibenzofuran (DBF) units to expand $\pi$-conjugation and narrow the energy gap ($E_g$), effectively steering the emission into the green region. To achieve finer modulation of the emission profile, rigid cyano moieties were incorporated para to the boron centers to exert tighter control over excited-state structural relaxation. Beyond positioning the emission within the pure-green region, this modification was expected to attenuate structural and vibronic relaxation, ultimately favoring the generation of an ultra-narrow pure-green emission profile.

To gain theoretical insight into how these progressive structural modifications regulate the electronic structure and excited-state characteristics, density functional theory (DFT) and time-dependent DFT (TD-DFT) calculations were performed using the B3LYP/6-31G(d,p) method. Considering that peripheral substituents exert little influence on the electronic structure, simplified models DBN-core, DBFBN-core, and DBFCN-core, excluding the dangling 2,6-dimethylphenyl substituents, were employed for the calculations [43–46]. These compounds preserve the characteristic MR distribution in their frontier molecular orbitals (FMOs), with the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) remaining spatially alternated, which underpins their narrowband emission (Figure 2 and Figure S19). More importantly, the stepwise structural modification from DBN-core to DBFBN-core and further to DBFCN-core continuously lowers the LUMO level and, together with the accompanying HOMO modulation, progressively narrows the energy gap ($E_g$) from 3.37 to 3.11 and ultimately to 3.02 eV. This evolution mainly arises from the $\pi$-conjugation extension introduced by the DBF group and the stronger electron-accepting character of the cyano substituents. Consistently, the computed singlet excited-state energy decreases accordingly, foreshadowing a continuous bathochromic shift of the emission. To further clarify the excited-state character, natural transition orbital (NTO) analysis was performed for the key emitter cores (Figure S20). The hole and particle distributions of DBN-core, DBFBN-core, and DBFCN-core are mainly localized within the same MR framework with substantial spatial overlap, confirming their MR-type short-range charge-transfer transition character rather than pronounced long-range charge-transfer separation [47, 48]. This preserved MR-type excited-state character provides an electronic-structure basis for realizing ultra-narrow emission.

To further elucidate how the FWHM evolves during the bathochromic shift, we evaluated the geometric displacement between the optimized ground state ($S_0$) and first singlet excited state ($S_1$) of DBN-core, DBFBN-core, and DBFCN-core, using the root-mean-square displacement (RMSD) as a quantitative descriptor. All emitters exhibit relatively small RMSD (0.0865 Å for DBN-core, 0.0833 Å for DBFBN-core, and 0.0588 Å for DBFCN-core), indicative of intrinsically weak excited-state structural relaxation (Figure 2a). Particularly noteworthy is that the transition from DBN-core to DBFBN-core produces only a marginal reduction in RMSD, suggesting that the fused-ring modification primarily drives the bathochromic shift while exerting minimal influence on excited-state geometric reorganization. In striking contrast, DBFCN-core displays a distinctly contracted RMSD, indicating that cyano substitution effectively suppresses structural relaxation, thereby providing a direct structural basis for its ultra-narrow emission bandwidth. Franck–Condon simulations of DBN-core, DBFBN-core, and DBFCN-core further reveal the corresponding evolution of the spectral bandwidths. As depicted in Figure 2b, the simulated spectrum remains in the blue region for DBN-core, with the maximum located at 456 nm and the FWHM limited to 0.078 eV, whereas DBFBN-core undergoes a pronounced bathochromic shift to 502 nm while retaining a similarly narrow bandwidth (0.075 eV). More strikingly, DBFCN-core exhibits a further bathochromic shifted emission peak at 513 nm, together with a substantially reduced bandwidth of 0.055 eV. These results reveal clear mechanistic differentiation that the fused-ring extension mainly modulates the emission energy, whereas cyano decoration plays a more decisive role in curbing excited-state structural relaxation and the associated vibronic broadening. As a result, DBFCN-core uniquely achieves a desirable synergy of bathochromic shift and spectral narrowing. To further unravel the origin of the bandwidth evolution, we analyzed the Huang–Rhys factor (HRFs) distributions together with the corresponding representative vibrational modes, employing the Molecular Material Properties Prediction Package (MOMAP) (Figure 2c,d) [49, 50]. All emitters deliver similar vibronic characteristics dominated by low-frequency modes, underscoring the critical role of excitation-induced twisting motions in shaping the emission envelope. For DBN-core, the principal contributions arise from modes at 19.65 and 53.28 $cm^{-1}$ with relatively large HRFs of 0.65 and 0.88. For DBFBN-core, the dominant coupling shifts to modes at 19.76 and 25.33 $cm^{-1}$, associated with HRFs of 0.54 and 1.01, indicating that ring fusion mainly reorganizes the relaxation pattern while preserving considerable low-frequency vibronic activity, which is consistent with the minimal changes in both RMSD and FWHM. By contrast, DBFCN-core exhibits a clear attenuation of the corresponding torsional modes, with the leading HRFs decreasing to 0.26 and 0.42 at 17.28 and 24.23 $cm^{-1}$, respectively. Collectively, the calculations unveil a clear synergy that rigid conjugation extension facilitates the emission redshift and cyano-decoration further effectively suppresses vibronic coupling. Such an interplay serves as a compelling theoretical foundation for the ultra-narrow pure-green emission inherent to the DBFCN-core.

### 2.2 | Synthesis and Characterization

Motivated by the computational results described above, we synthesized DBN, DBFBN, and DBFCN based on the model

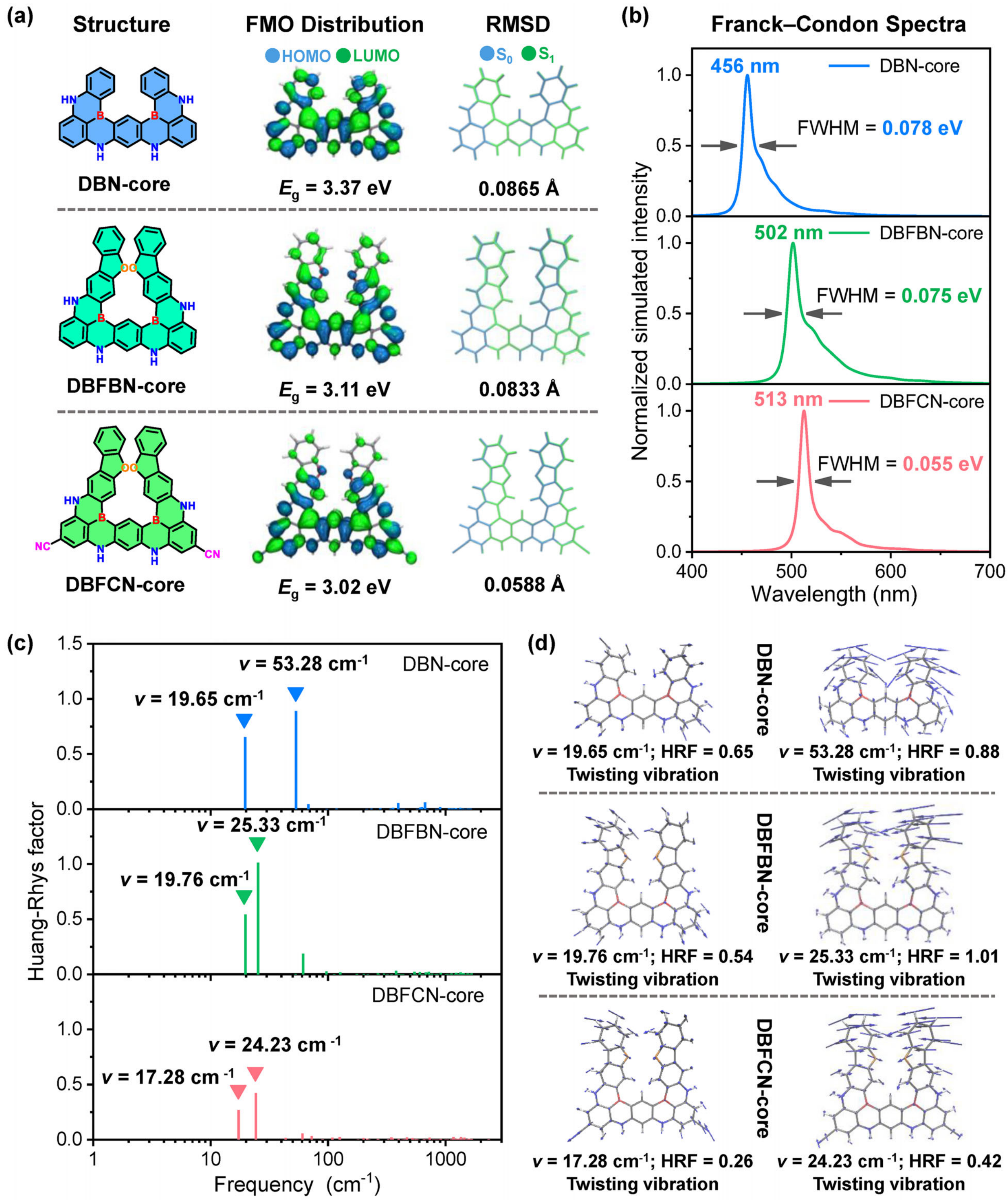


**FIGURE 2** | Theoretical analysis of DBN-core, DBFBN-core, and DBFCN-core. (a) The molecular structure, frontier molecular orbital distributions, and root-mean-square-displacement (RMSD) values. (b) Emission spectra for the $S_1 \rightarrow S_0$ transition simulated on the basis of Franck–Condon analysis. (c) Huang-Rhys factor versus frequency plots for the $S_1 \rightarrow S_0$ transition. (d) Representative vibrational modes involved in the $S_1 \rightarrow S_0$ transition of DBN-core, DBFBN-core, and DBFCN-core.

compounds. The synthesis procedures for target emitters are shown in Schemes S1 and S2. The pivotal chlorinated intermediates, DBNCl and DBFCl, were prepared through a regioselective one-step borylation reaction [19, 51]. These intermediates then served as common precursors for the target emitters, which were accessed through either dechlorination or palladium-catalyzed cyano substitution of the C–Cl bonds. After purification by column chromatography and vacuum sublimation, structural characterization of all target compounds was accomplished by $^{1}H$, $^{13}C$ nuclear magnetic resonance (NMR) spectroscopy and high-resolution mass spectrometry (HRMS) analysis (Figures S2–S17).

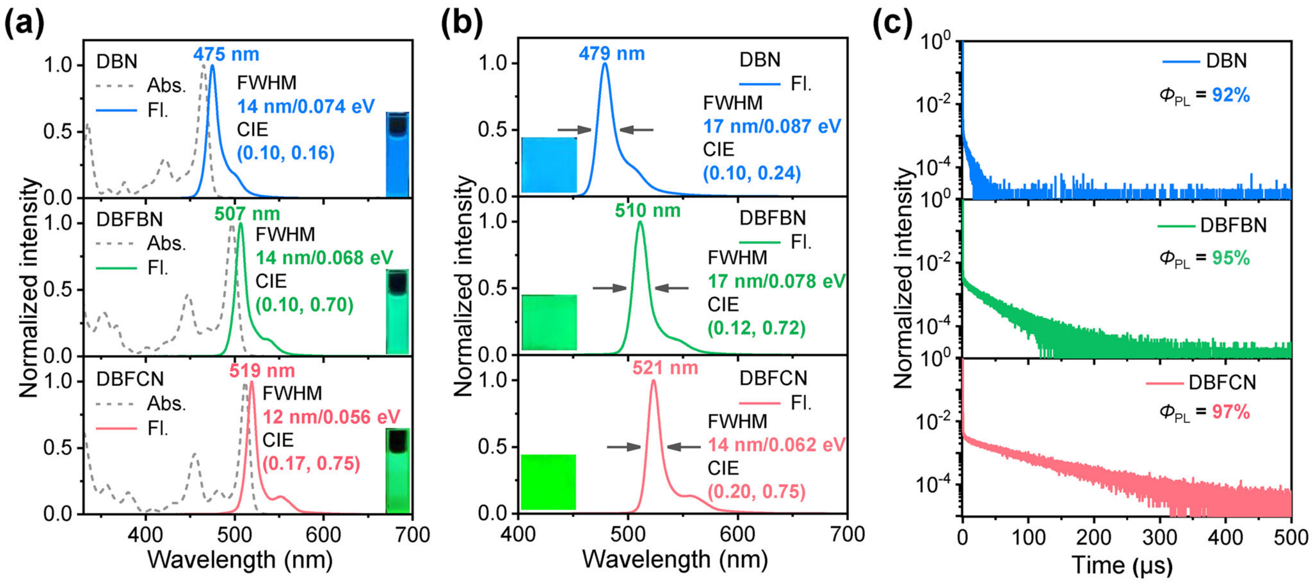


**FIGURE 3** | Photophysical properties of emitters. (a) Solution-phase absorption and steady-state emission spectra of DBN, DBFBN, and DBFCN measured in dilute toluene solution ($1.0 \times 10^{-5}$ M). (b) Steady-state emission spectra of DBN, DBFBN, and DBFCN in doped films. (c) Time-resolved PL decay curves of the doped films.

A single crystal of DBFCN amenable to single-crystal X-ray diffraction was obtained through slow ethanol diffusion into a dichloromethane solution, providing definitive structural information (Figure S18 and Table S1). The structural analysis reveals that the central MR core undergoes a slightly butterfly-like conformation, while the pendent 2,6-dimethylphenyl groups adopt an almost orthogonal arrangement with respect to the skeleton, with dihedral angles of 87.1–89.9°. In addition, the upper aryl fragment remains significantly twisted, showing a dihedral angle of 49.5°, further increasing the overall conformational nonplanarity. Such a highly twisted geometry generates substantial steric shielding around the intrinsically delicate MR core and effectively disfavors close molecular packing. Under this sterically encumbered protection, detrimental intermolecular interactions are efficiently suppressed in the doped films. Consequently, excimer formation and aggregation-caused quenching are expected to be alleviated, which is beneficial for preserving the ultra-narrow green emission of DBFCN and may further contribute to improved device durability.

## 2.3 | Photophysical Properties

The solution-phase photophysical characteristics of DBN, DBFBN, and DBFCN were evaluated in dilute toluene solutions. DBN, DBFBN, and DBFCN exhibit intense absorption maximum at 465, 497, and 512 nm, respectively, arising from the short-range charge-transfer transitions characteristic of multi-resonance compounds. Consistent with corresponding computational results, the PL spectra show a clear bathochromic progression from DBN to DBFCN while preserving extremely narrow emission profiles, with peaks at 475, 507, and 519 nm and corresponding FWHMs of 14 nm (0.074 eV) for DBN, 14 nm (0.068 eV) for DBFBN, and 12 nm (0.056 eV) for DBFCN, respectively (Figure 3a and Table 1). Stokes shifts further corroborate the theoretical prediction of reduced structural relaxation. DBN and DBFBN both exhibit Stokes shifts of 10 nm, whereas DBFCN shows a smaller value of only 7 nm, consistent with its more effectively suppressed structural relaxation between the $S_1$ and $S_0$ states. Notably, DBFCN combines green emission with the narrowest bandwidth in the series, placing it among the sharpest pure-green MR-TADF emitters reported so far (Table S2 and Figure S21). Solvent-dependent PL spectra further show limited emission shifts and well-maintained narrow FWHMs in solvents with different polarities, especially for DBFCN, manifesting its MR-type short-range charge-transfer emissive character (Figure S22 and Table S3). On the basis of the fluorescence and phosphorescence spectra at 77 K, the singlet-triplet energy gap ($\Delta E_{ST}$) values were determined as 0.09 eV for DBN, 0.10 eV for DBFBN, and 0.12 eV for DBFCN (Figure S23). Such small $\Delta E_{ST}$s are favorable for efficient reverse intersystem crossing, a prerequisite for achieving superior-performance TADF emission.

Solid-state photophysical characteristics were examined in vacuum-deposited films prepared with 3 wt.% emitter dispersed in DMIC-TRZ [52]. All emitters retain excellent emissive characteristics, with photoluminescence quantum yields ($\Phi_{PL}$s) of 92%, 95%, and 97% for DBN, DBFBN, and DBFCN, respectively. The nearly unity $\Phi_{PL}$ value of the DBFCN-doped film is consistent with its rigidified molecular framework and effectively suppressed structural relaxation, which together minimize nonradiative deactivation. The emission peaks of DBN, DBFBN, and DBFCN are located at 479, 510, and 521 nm, respectively, showing only slight red shifts relative to those in dilute toluene. Despite the increased polarity and intermolecular effects in the doped films, the spectra remain exceptionally narrow, with FWHMs of 17 nm (0.087 eV), 17 nm (0.078 eV), and 14 nm (0.062 eV), respectively. Significantly, the combination of a well-positioned green emission peak and an exceptionally compressed bandwidth enables DBFCN to attain a highly desirable $CIE_y$ value of 0.75, closely matching the BT.2020 green

**TABLE 1** | Photophysical properties of DBN, DBFBN, and DBFCN.

| Emitter | $\lambda_{abs}$[a] [nm] | $\lambda_{em}$[a] [nm] | FWHM[a] [nm/eV] | $\Delta E_{ST}$[b] [eV] | $\Phi_{PL}$[c] [%] | $\tau_{PF}$[c] [ns] | $\tau_{DF}$[c] [µs] | $k_{r,S}$[c] [$10^7$ $s^{-1}$] | $k_{nr,S}$[c] [$10^6$ $s^{-1}$] | $k_{RISC}$[c] [$10^5$ $s^{-1}$] |
|---|---|---|---|---|---|---|---|---|---|---|
| DBN | 465 | 475 | 14/0.074 | 0.09 | 92 | 4.2 | 6.2 | 16.9 | 14.7 | 2.0 |
| DBFBN | 497 | 507 | 14/0.068 | 0.10 | 95 | 5.9 | 34.1 | 4.2 | 2.2 | 1.1 |
| DBFCN | 512 | 519 | 12/0.056 | 0.12 | 97 | 4.3 | 64.6 | 3.7 | 1.2 | 0.9 |

[a]Recorded in dilute toluene ($1 \times 10^{-5}$ M).
[b]$\Delta E_{ST} = E_{S1}$ - $E_{T1}$, where $E_{S1}$ and $E_{T1}$ were derived from the onsets of fluorescence and phosphorescence at 77 K in toluene.
[c]Measured in 3 wt% doped films.

**TABLE 2** | Performance summary of the OLEDs.

| Emitter | $V_{on}$[a] [V] | $\lambda_{EL}$[b] [nm] | FWHM[b] [nm/eV] | $L_{max}$ [cd $m^{-2}$] | $CE_{max}$ [cd $A^{-1}$] | $PE_{max}$ [lm $W^{-1}$] | EQE[c] [%] | CIE[b] (x, y) |
|---|---|---|---|---|---|---|---|---|
| DBFBN | 2.4 | 510 | 17/0.079 | 84893 | 103.3 | 125.5 | 34.3/28.6/20.5 | (0.11, 0.72) |
| DBFCN | 2.4 | 521 | 13/0.060 | 71683 | 147.9 | 193.6 | 38.5/34.4/21.2 | (0.18, 0.75) |
| DBFCN-**HF** | 2.8 | 521 | 14/0.062 | 157487 | 150.3 | 168.7 | 38.3/38.2/35.1 | (0.21, 0.74) |
| DBFCN-**TE** | 2.4 | 522 | 13/0.060 | 50874 | 232.5 | 304.2 | 58.9/58.8/57.6 | (0.14, 0.79) |

[a]Turn-on voltage at 1 cd $m^{-2}$.
[b]Recorded at 1000 cd $m^{-2}$.
[c]EQE at maximum, 100 and 1000 cd $m^{-2}$.

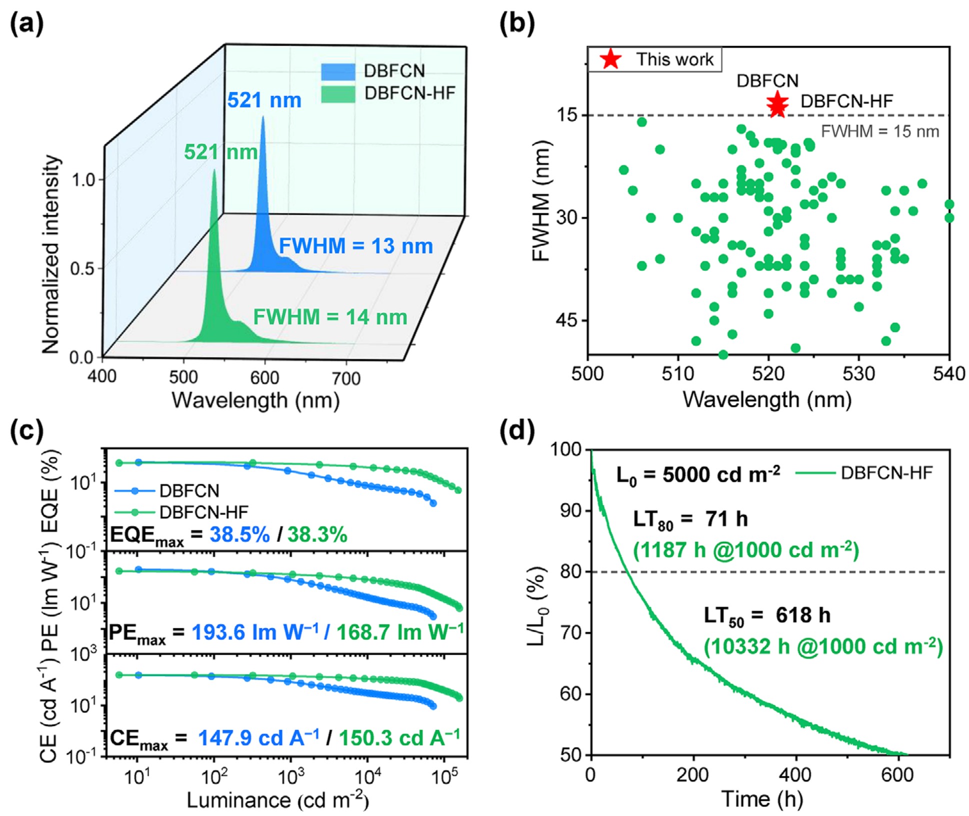


**FIGURE 4** | Electroluminescent performance of the bottom-emitting green OLEDs. (a) Electroluminescence spectra of DBFCN and DBFCN-HF recorded at 1000 cd m$^{-2}$. (b) Reported green MR-TADF materials with FWHM ≤ 50 nm in bottom-emitting OLEDs. (c) Luminance-dependent EQE, PE, and CE characteristics of DBFCN and DBFCN-HF. (d) Device stability of the DBFCN-HF was evaluated at 5000 cd m$^{-2}$.

target. These results confirm that the ultra-narrow emission character can be well preserved in the solid state.

Room-temperature transient photoluminescence measurements of the 3 wt.% doped films reveal distinct TADF behavior (Figure 3c and Figure S24). Each film shows a characteristic biexponential decay, comprising a nanosecond prompt component ($\tau_{PF}$) and a microsecond delayed component ($\tau_{DF}$). The corresponding prompt/delayed lifetimes are 4.2 ns/6.2 µs for DBN, 5.9 ns/34.1 µs for DBFBN, and 4.3 ns/64.6 µs for DBFCN, respectively, $k_{RISC}$ gradually decreases from DBN to DBFBN and DBFCN, with values of 2.0, 1.1, and 0.9 × 10$^5$ s$^{-1}$, respectively. This trend can be mainly attributed to the enlarged $\Delta E_{ST}$ (Table 1 and Figure S23), which slows the RISC process and prolong $\tau_{DF}$ [53]. Nevertheless, DBFCN still maintains a moderate $k_{RISC}$, a high $\Phi_{PL}$, and suppressed nonradiative decay, thereby ensuring its excellent luminescent performance (Table S4). Angle-resolved PL measurements reveal nearly perfect horizontal dipole orientation ($\Theta_{//}$) for these emitters in doped films (Figure S25). Such highly preferential horizontal alignment should markedly increase optical out-coupling efficiency ($\eta_{out}$), thereby providing a photophysical basis for the high external quantum efficiencies (EQEs) of the devices [54–56]. Before device preparation, the energy levels of DBFBN and DBFCN were examined by cyclic voltammetry and optical-bandgap analysis. Their HOMO/LUMO values were calculated to be −5.09/−2.66 eV for DBFBN and −5.38/−3.02 eV for DBFCN, respectively (Figure S26 and Table S5). Both compounds also exhibit good thermal stability, with $T_d$ (decomposition temperature) values at 5% mass loss of 406°C for DBFBN and 470°C for DBFCN, respectively (Figure S27), which is highly advantageous for vacuum-deposited OLEDs.

## 2.4 | OLEDs Performances

To probe the electroluminescence (EL) characteristics of DBFBN and DBFCN, non-sensitized OLEDs were constructed using the following multilayer structure of ITO/HATCN (5 nm)/BPBPA (50 nm)/BiCar (15 nm)/host: 3 wt% emitters (55 nm)/POT2T (20 nm)/ANT-BIZ (30 nm)/Liq (2 nm)/Al (100 nm). Within this device architecture, ITO and Al serve as the anode and cathode, respectively. HATCN and Liq were incorporated to assist charge injection, hole and electron transport were supported by BPBPA and ANT-BIZ, and POT2T together with BiCar provided hole and electron blocking around the emitting layer. DMIC-TRZ served as the host in view of its favorable energy alignment with adjacent layers, as well as its ambipolar transport properties and small $\Delta E_{ST}$, which collectively support balanced charge injection and effective exciton utilization. Table 2 compiles the principal device parameters, whereas Figure 4 and Figures S28–S32 display the energy-level diagram, the molecular structures of the device materials, and the corresponding electroluminescence behavior.

The DBFBN-based device delivers narrowband green electroluminescence maximum at 510 nm with a relatively small FWHM of 17 nm (0.079 eV). Along with its good color purity (CIE$_y$ = 0.72), accompanied by a maximum external quantum efficiency (EQE$_{max}$) of 34.3%, maximum current efficiencies (CE$_{max}$) of

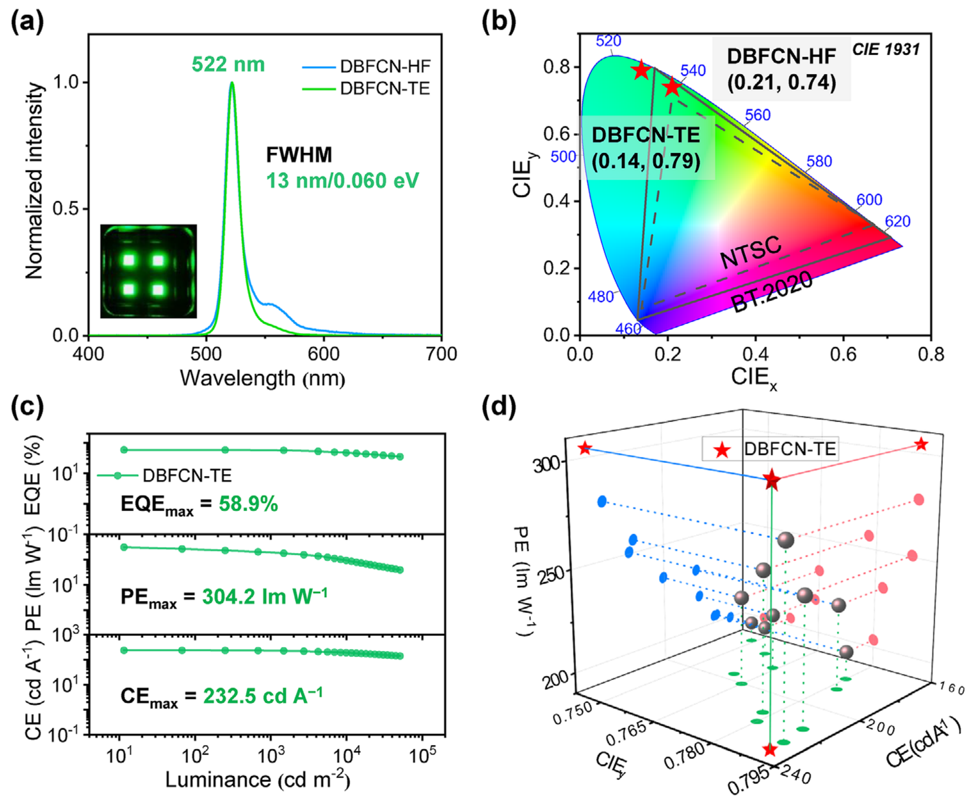


**FIGURE 5** | Electroluminescent performance of the top-emitting green OLED. (a) Electroluminescence spectra of the DBFCN-HF and DBFCN-TE measured at 1000 cd m$^{-2}$. (b) CIE coordinates measured at 1000 cd m$^{-2}$. (c) Variation of EQE, PE, and CE with luminance for DBFCN-TE. (d) 3D scatter plot of PE, CE, and CIE$_y$ for the DBFCN-TE device (red stars) and literature-reported TE devices (grey spheres).

103.3 cd A$^{-1}$, and maximum power efficiencies (PE$_{max}$) of 125.5 lm W$^{-1}$ (Figure S29). While these results already position DBFBN among high-performance green emitters with excellent color purity, the DBFCN-based device pushes the performance into a distinctly more advanced regime. Specifically, the DBFCN-based OLED exhibits a sharply saturated pure-green electroluminescence, characterized by an emission maximum at 521 nm and a remarkably narrow bandwidth of 13 nm (0.060 eV), which stands among the narrowest bottom-emitting pure-green MR-TADF OLEDs reported to date (Table S6 and Figure 4b). The sharp emission spectrum and highly precise maximum emission wavelength give rise to CIE coordinates of (0.18, 0.75), very close to the BT.2020 pure-green target (Figure S30). In addition, the device delivers exceptional EL performance, with the EQE$_{max}$ of 38.5%, alongside CE$_{max}$ of 147.9 cd A$^{-1}$ and PE$_{max}$ of 193.6 lm W$^{-1}$ (Figure 4c). The concurrent realization of an ultra-narrow FWHM of 13 nm, EQE$_{max}$ approaching 40%, and PE$_{max}$ approaching 200 lm W$^{-1}$ underscores a remarkably balanced convergence of color purity and EL efficiency, positioning it as one of the most compelling pure-green binary OLEDs reported so far. A phosphorescent sensitizer was further introduced to construct a ternary device, which maintains high efficiency and excellent color purity while delivering markedly suppressed efficiency roll-off and enhanced operational lifetime (Figure S32). Under constant-current operation with the initial brightness of 5000 cd m$^{-2}$, DBFCN-HF affords LT$_{50}$ of 618 h, corresponding to an extrapolated lifetime of 10332 h (LT$_{50}$) at a luminance of 1000 cd m$^{-2}$. Collectively, these findings reveal that the DBFCN-based sensitized OLED achieves a good balance among excellent EL efficiency, color purity, and good device stability, placing it among the most advanced pure-green MR-TADF OLEDs (Table S6).

Encouraged by the excellent performance and highly concentrated emission of the bottom-emitting devices, we further advanced the device engineering to a top-emitting (TE) configuration in order to exploit the intrinsic ultra-narrow emission of DBFCN under a cavity-controlled optical environment. The TE-OLED was constructed with the architecture: ITO/HATCN (5 nm)/BPBPA (142 nm)/BiCar (15 nm)/EML (35 nm)/ANT-BIZ (45 nm)/Yb (3 nm)/Ag (12 nm)/BPBPA (75 nm). In this architecture, the semitransparent top electrode together with the reflective interface establishes a well-defined optical microcavity, which further confines the EL spectrum and suppresses the long-wavelength vibronic tail. As illustrated in Figure 5 and Figure S34, DBFCN-TE delivers highly purified green electroluminescence, with the emission maximum located at 522 nm and an ultra-narrow FWHM of 13 nm (0.060 eV), and CIE coordinates of (0.14, 0.79), almost reaching the BT.2020 green target (Figure 5b). The DBFCN-TE device exhibits excellent spectral stability over a broad driving-voltage range of 2.4–10.0 V (Figure S33), with the ultra-narrow pure-green EL profile remaining essentially unchanged, suggesting stable exciton formation and efficient energy transfer within the emitting layer under different operating conditions. Meanwhile, the device achieves a remarkable EQE$_{max}$ of 58.9%, CE$_{max}$ of 232.5 cd A$^{-1}$, and PE$_{max}$ of 304.2 lm W$^{-1}$ without any external light-extraction structure, placing DBFCN-TE among the best-performing top-emitting pure-green OLEDs reported to date (Figure 5d and

Tables S7 and S8). Notably, the electroluminescence spectrum of the TE-OLED closely resembles that of its bottom-emitting counterpart, indicating that the emission remains predominantly governed by the intrinsic narrowband character of DBFCN, while the microcavity mainly serves to further purify the spectrum. The coexistence of this nearly unchanged spectral profile with very high device efficiency suggests that cavity-induced narrowing is achieved without a severe outcoupling penalty. This finding underscores the exceptional promise of intrinsically ultra-narrow emitters for high-performance top-emitting OLEDs.

## 3 | Conclusions

In brief, we have constructed an ultra-narrow pure-green MR-TADF emitter, DBFCN, through a synergistic molecular design that combines fusion extension with cyano decoration on a *meta*-diboron framework. Theoretical and experimental analyses consistently reveal a clear functional differentiation between these two structural modifications: fusion extension primarily drives the bathochromic shift, whereas cyano decoration more effectively suppresses excited-state structural relaxation and low-frequency vibronic coupling. Their synergy thus enables the unusual yet highly desirable coexistence of red-shift and further spectral narrowing. As a result, DBFCN exhibits intrinsically ultra-narrow pure-green PL at 519 nm with an FWHM limited to 12 nm in solution. More importantly, the bottom-emitting OLED delivers sharply saturated pure-green electroluminescence at 521 nm with a very narrow FWHM of only 13 nm, together with CIE coordinates of (0.18, 0.75), $EQE_{max}$ approaching 40%, and $PE_{max}$ almost reaching 200 lm $W^{-1}$, placing it among the narrowest and most compelling bottom-emitting pure-green MR-TADF OLEDs reported so far. The practical value of such intrinsically ultra-narrow emission is even more compellingly demonstrated in top-emitting devices. Unlike conventional broadband emission systems, where microcavity modulation often relies on passive spectral filtering at the expense of substantial off-resonance photon loss, DBFCN remains highly compatible with the microcavity architecture, enabling the FWHM to remain essentially unchanged while simultaneously enhancing color purity and device efficiency. Consequently, the device delivers a high $EQE_{max}$ approaching 60%, $CE_{max}$ of 232.5 cd $A^{-1}$, and $PE_{max}$ exceeding 300 lm $W^{-1}$, accompanied by excellent color purity approximately meeting the BT.2020 green standard. These results establish synergistic fusion extension and cyano decoration as a viable principle for realizing superior-performance pure-green MR-TADF emitters, while underscoring the exceptional promise of intrinsically ultra-narrow emitters for fully exploiting microcavity modulation in top-emitting OLEDs.


### Acknowledgements

This work was supported by the National Natural Science Foundation of China (22301187 and 52130308), the Guangdong Basic and Applied Basic Research Foundation (2026A1515010023), the Shenzhen Science and Technology Program (JCYJ20220818095816036 and ZDSYS20210623091813040), the Research Team Cultivation Program of Shenzhen University (2023DFT004) and Scientific Foundation for Youth Scholars of Shenzhen University (868-000001033366). The authors also thank the Instrumental Analysis Center of Shenzhen University for analytical support.


### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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### Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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